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Oefeningenreeks 4A nr. 44.1

Los volgende integraal op a.d.h.v Fuss: $I = \int \frac{5x^2+3}{x^3+x} dx$

$$I = \int \frac{5x^2+3}{x^3+x} dx = \int \frac{5x^2+3}{x(x^2+1)} dx$$

Neem $A, B, C \in \mathbb{R}$ zodat:

$$I = \int \left(\frac{A}{x} + \frac{Bi+C}{x^2+1} \right) dx$$

met:

$$A = \left. \frac{5x^2+3}{x^2+1} \right|_{x=0} = 3$$

$$Bi + C = \left. \frac{5x^2+3}{x} \right|_{x=i} = 2i + 0$$

$$\Rightarrow (B, C) = (2, 0)$$

$$\Rightarrow I = 3 \int \frac{1}{x} dx + \int \frac{2x}{x^2+1} dx$$

$$\Rightarrow I = 3 \ln |x| + C + \int \frac{2x}{x^2+1} dx$$

Stel: $t = x^2 + 1$

$$\Rightarrow dt = dx^2 + 1 = 2xdx$$

$$\Rightarrow \int \frac{2x}{x^2+1} dx = \int \frac{1}{t} dt$$

$$\Rightarrow I = 3 \ln |x| + \ln |t| + C$$

$$\Rightarrow I = 3 \ln |x| + \ln |x^2+1| + C$$

References